Proposition 5.21.2.

$$S_\lambda(1, z, z^2, \dots, z^{N-1}) = \prod_{1 \le i < j \le N} \frac{z^{\lambda_i - i} - z^{\lambda_j - j}}{z^{-i} - z^{-j}}.$$

Therefore,

$$S_\lambda(1, \dots, 1) = \prod_{1 \le i < j \le N} \frac{\lambda_i - \lambda_j + j - i}{j - i}.$$

Proof. First, $D_\lambda(1, z, \dots, z^{N-1})$ is a Vandermonde determinant evaluated at $(z^{\lambda_i + N - i})_{1 \le i \le N}$, so it equals $\prod_{i<j}(z^{\lambda_i + N - i} - z^{\lambda_j + N - j})$. Dividing by the same formula with $\lambda = 0$ yields the formula for $S_\lambda(1, z, \dots, z^{N-1})$. Now take $\lim_{z \to 1}$ and apply L'Hôpital's rule. $\square$

5.22. The characters of L_λ

Proposition 5.21.1 allows us to calculate the characters of the representations L_λ.

Namely, let $\dim V = N$, let $g \in GL(V)$, and let $x_1, \dots, x_N$ be the eigenvalues of g on V. To compute the character $\chi_{L_\lambda}(g)$ as a function of $x_1, \dots, x_N$, let us calculate $\mathrm{Tr}_{V^{\otimes n}}(g^{\otimes n} s)$, where $s \in S_n$. If $s \in C_{\mathbf{i}}$, we easily get that this trace equals

$$\prod_m \mathrm{Tr}(g^m)^{i_m} = \prod_m H_m(x)^{i_m}.$$

On the other hand, by the Schur-Weyl duality

$$\mathrm{Tr}_{V^{\otimes n}}(g^{\otimes n} s) = \sum_\lambda \chi_\lambda(C_{\mathbf{i}}) \mathrm{Tr}_{L_\lambda}(g).$$

Comparing this to Proposition 5.21.1 and using linear independence of columns of the character table of S_n, we obtain

Theorem 5.22.1 (Weyl character formula)**.** *The representation L_λ is zero if and only if $N < p$, where p is the number of parts of λ. If $N \ge p$, the character of L_λ is the Schur polynomial $S_\lambda(x)$. Therefore, the dimension of L_λ is given by the formula*

$$\dim L_\lambda = \prod_{1 \le i < j \le N} \frac{\lambda_i - \lambda_j + j - i}{j - i}.$$